Sustainability and Human Impact of Energy-Aware Decision Making in Cloud Data Centers

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Abstract

Cloud data centers are a rapidly growing source of global electricity consumption, yet energy-aware operational strategies are predominantly evaluated on carbon footprint alone, neglecting water footprint, user fairness, and community-level impacts. This thesis develops an interpretable framework that jointly quantifies the sustainability outcomes and human impact of energy-aware strategies using fine-grained node- and VM-level operational data from the IFCA Cloud. The framework uses interpretable machine learning models to forecast energy consumption and simulate strategy trade-offs, providing transparent decision support for cloud operators balancing sustainability and fairness trade-offs. The corresponding repository: https://github.com/bchiusano/Thesis

1 Introduction

The rapid growth of cloud computing and artificial intelligence has caused data centers (DC) to currently account for 1-2% of global electricity consumption, with projections suggesting an increase to 9% by 2030, despite significant improvements in carbon-efficient implementations. A major improvement in energy consumption over the last few years is attributed to the shift to cloud computing, where cloud vendors distribute physical resource utilization through virtual units. These are sold to users and organisations by the second, hour, or per request. Even though the physical equipment still exists, resources are better allocated, and usage is optimized. Simultaneously, however, the cloud infrastructure makes it more challenging for customers to be aware of the environmental cost of computation. Overall, there is a lack of transparency regarding the reported energy usage from DCs, leading to imprecise estimations of the energy management of these infrastructures.

An even less reported and understood environmental consequence of data center usage is its water consumption, thus making this a constantly developing research field. While energy-aware operational strategies, such as workload placement and job scheduling techniques, have been widely studied as mechanisms to reduce the carbon footprint (CF) of cloud data centers (CDC), the water footprint of these operations remains unaddressed. Cooling and Electricity generation for CDCs are strongly related to water consumption; the CF and WF of both the centers and their energy-efficient operational strategies should not continue to be studied independently.

While energy-aware operational strategies have been seen to reduce the energy use and CF of cloud data centers, there is less focus on how these implementations may impact users differently, for example, affecting delays or resource availability. This leads to a study on fairness, regarding whether these strategies impose a disproportionate burden on specific user classes.

These gaps will be addressed using the fine-grained node- and VM-level energy dataset from the Instituto de Fisica de Cantabria (IFCA) Cloud [2], as well as carbon and water consumption measurements from the ENTSO-E transparency platform and the Wattnet tool [3, 8]. By implementing an interpretable energy forecasting model, this work will aim to quantify the trade-offs of different operational strategies, between sustainability outcomes (CF and WF reduction), the human ethical impact, and fairness across user group. The project begins with baseline interpretable models and later extends to state-of-the-art temporal architectures (LSTM/GRU). The outcome will serve to provide transparent human-centered decision support that balances sustainability with user needs.

The preliminary research question that will be answered is:
(RQ) *To what extent do energy-aware operational strategies in cloud data centers influence the trade-off between environmental sustainability and user impact, and how can these trade-offs be quantified transparently?*
This is addressed through the following subquestions:

- **(RQ1)** *How accurately can interpretable machine learning models forecast node and VM-level energy consumption to support strategy evaluation?*
- **(RQ2)** *What is the quantitative impact of energy-aware operational strategies on carbon footprint and water footprint reduction?*
- **(RQ3)** *To what extent do sustainability-driven operational strategies disproportionately affect specific user groups?*
- **(RQ4)** *How can an interpretable decision support framework improve transparency for cloud operators evaluating sustainability and fairness trade-offs?*

2 Related Work

2.1 Energy Consumption

Dependence on digital services and the use of artificial intelligence are among the factors causing electricity consumption to grow at an unprecedented rate. Data centers today consume around 3% of the world's electricity resources, and it is expected to reach up to 9% by 2030 [5]. The work of Orozco et al. presents a forecast of the electricity and water footprint of digital data services in Europe until 2030, demonstrating that 273.4 to 820.1 million cubic meters of water and from 56.3 to 169 Terawatt-hour of electricity will be consumed yearly for internet usage, meaning that in 4 years the average European user could be using more water in internet than the amount needed for drinking. Nevertheless, energy efficiency of data centers is constantly increasing to the point that an increase in computing workload of 550% from 2010 to 2018 has resulted only in an increase of energy demand of 6% [4].

One of the main contributions to which this work will draw particular attention is an open-source tool called Wattnet. This project was also developed by one of the authors of the IFCA Cloud Dataset. Wattnet is designed to provide a joint assessment of both the CF and WF of electricity consumption. Two claims are made regarding the gap in previous works: that cross-border electricity flows are largely overlooked and that the water impacts of electricity generation are usually neglected. Thus, their work accounts for the spatial and temporal dynamics of electricity generation and highlights cross-border energy flows in Europe, aiming to facilitate informed decision-making for energy-aware operations of DCs and enhancing transparency for end users. Their flow-tracing methodology allows to calculate the energy mix composition of a country or region taking into account not only the country's electricity consumption but the fluxes between neighboring countries. Considering that the electricity consumed in a given European electric grid is composed not only of the electricity generated in the region where this grid is located but also of the electricity that this region imports. Ignoring electricity flows, imports and exports, can lead to a systematic under- or over estimation of both CF and WF. Results from the wattnett study highlight the strong interdependence between energy and water systems, especially in DCs where both carbon emissions and water use occur largely off-site through electricity consumption, thus in future work CF and WF should not be analysed independently. Moreover they show that countries that rely on electricity from hydropower-intensive regions are particularly affected, such as Spain, where IFCA cloud is located. [8].

2.2 Water Consumption

In contrast with power consumption, which can only occur on-site, water consumption in DCs can occur both directly and indirectly. Direct or On-site water consumption refers to the water used for the cooling of the rooms hosting the hardware using in DCs. This is a necessary step to prevent equipment failure as temperatures rise. In cooler regions, the cold air from the environment can reduce the amount of artificial cooling required [9]. In terms of transparency, on-site water consumption can be monitored, but it's not often reported. The use of water for cooling has the advantage of reducing electricity consumption but still imposes other environmental changes and contributes to the overall WF of a CDC [8]. Moreover, on-site water consumption is particularly controversial since more data centers are opening in areas where water is scarce [4].

On the other hand, indirect water consumption occurs off-site and depends on the electricity source used to power the infrastructure. The water consumption for electricity production can vary greatly depending on the source. Thermal power plants and hydropower plants have considerably a higher water consumption than renewables like solar photovoltaic and wind. The assessment of off-site water usage requires having knowledge about the energy mix in the grid that supplies the DC with electricity [8]. Another distinction that needs to be made is between Local and Global Carbon and Water Footprints. Local footprints are calculated using only locally generated electricity while Global footprints account for electricity imports and export. Optimization of energy and water use can be a trade-off in some cases reducing the processing performance of datacenters The results reveal that a shift to a decarbonized system can be detrimental to water resources if it mainly relies on hydropower from reservoirs, while other low-carbon intensive energy sources like solar and wind do not have an impact on water systems. Optimising both the CF and the WF at the same time can be challenging. The case of reservoir-based hydropower must be given particular attention when assessing the CF and WF of electricity generation. In contrast to most other generation technologies, whose operational environment impacts are directly linked to electricity production, hydropower reservoirs exert a continuous impact regardless of whether electricity is being generated or consumed. Moreover, DCs can make use of different systems. Closed-loop systems reuse the same water, reducing new water input but still facing evaporation losses while open-loop systems continuously draw water.

2.3 Energy Efficiency Metrics

The most widely used indicator for assessing the sustainability of electricity use is the Carbon Footprint (CF), as it quantifies the amount of greenhouse gases (GHG) released to the atmosphere. This is expressed as CO_2-equivalent mass per kilowatt-hour (kWh) consumed.

GHG Protocol. The GHG Protocol is a corporate standard designed to provide guidance to companies and organisations regarding GHG emissions. The protocol categorizes the emissions into three different scopes. Scope 1 deals with direct GHG emissions that occur from sources that are owned or controlled by a company. Scope 2 accounts for emissions from the generation of purchased electricity consumed, which physically occur at the facility where electricity is generated. Companies are expected to report on scopes 1 and 2. Finally, Scope 3 is an optional reporting of all other indirect emissions, which occur as a consequence of the activities, but are not controlled, of a company [10]. While the protocol establishes how to calculate the CF of electricity consumption, the Water Footprint (WF) can also be calculated by using the water use factors [8].
The work of Safari et al. is a systematic review of 25 different energy efficiency metrics, assessing their relevance, gaps, and applicability in modern CDC environments [11]. These are classified into two groups: IT-related and non-IT-related metrics.

IT-Related Metrics. IT-related metrics are defined as the measurements that assess how effectively IT resources utilize energy to perform computational tasks. These focus on the ratio of computational output to energy consumed. Table 1 displays all of the CDC energy efficiency IT-related metrics.

Non-IT-Related Metrics. Non-IT-related metrics evaluate the proportion of energy used by non-IT systems relative to total energy consumption. Non-IT systems entail all supporting infrastructure of a CDC that contribute to power consumption, such as IT rooms and equipment, HVAC (Heating, Ventilation, and Air Conditioning) systems, and facility security, lighting, and offices. A complete

Metric	Formulation	Definition
APC	$\frac{\text{Total IT Power}}{\text{Time}}$	IT power trends
CPE	$\frac{\text{Compute Output}}{\text{IT Power}}$	IT computational efficiency
DWPE	$\frac{\text{Workload}}{\text{IT Power}}$	Workload efficiency at IT level
EWR	$\frac{\text{IT Energy}}{\text{Work Output}}$	Energy cost of work
ITEE	$\frac{\text{Rated Performance}}{\text{Rated Power}}$	Hardware efficiency
OSWE	$\frac{\text{Workload}}{\text{Total System Energy}}$	System-level workload efficiency
PpW	$\frac{\text{Performance}}{\text{Power}}$	IT hardware efficiency
ScE	$\frac{\text{Output}}{\text{Server Power}}$	Server-level efficiency
SPUE	$\frac{\text{Total Server Energy}}{\text{Compute Energy}}$	Server-specific efficiency
SWaP	$\frac{\text{Performance}}{\text{Space} \times \text{Power}}$	Space-power performance balance

Table 1: IT-Related efficiency metrics

outline of non-IT-related metrics can be found in detail in the published systematic review [11]. However, as this paper will mainly focus on IT-related measurements, only Power Usage Effectiveness (PUE) will be addressed as it is the most widely adopted metric to evaluate CDC energy performance. PUE (Eq. 1) calculates the ratio of the data center total energy consumption to information technology equipment energy consumption (such as cooling) within a continuous 12-month period.

$$PUE = \frac{\text{Data Center Total Energy Consumption}}{\text{ICT Equipment Energy Consumption}} \tag{1}$$

Even though PUE is one of the more popular metrics used to communicate data center energy efficiency, it has been criticized for only conveying an understanding of the minimum possible energy use. The work of Yuventi and Mehdizadeh proposes using energy-based metrics or average PUE over a significant time period to better understand the energy efficiency of a DC.[13]. Moreover, the implementation of PUE across the industry is not standardized. In 2023, a study conducted by Isazadeh et al. found that the PUE calculations reported by 122 data centers varied up to 0.4 points due to inconsistent measurement methodologies [6]. Furthermore, PUE excludes software-related inefficiencies, dynamic workloads, and grid-level CO_2 intensity. Server Power Usage Effectiveness (SPUE), described in Table 1, is a proposed alternative which addresses energy use at more granular levels; however faces challenges in standardization and integration [11].

Water-related Metrics. Within the systematic review of energy efficiency metrics, there is no mention of water usage efficiency metrics for CDCs. Water Usage Effectiveness (WUE) was introduced by The Green Grid [12]. WUE is a site-based metric that assesses the water used on-site for humidification, on-site energy production, and cooling of the DC, defined by Eq 2.

$$WUE = \frac{\text{Annual Site Water Usage}}{\text{ICT Equipment Energy}} \tag{2}$$

WUE, however, only accounts for on-site water consumption, thus another proposed metric is WUE_{source} [9]. WUE_{source} is a source-based metrics that includes water used both on-site and off-site in the production of the energy used on-site [12]. The metric adds the measure of water used at the power-generation source to the water used on-site, and is defined by Eq 3.

$$WUE_{source} = \frac{\text{Annual Source Energy Water Usage + Annual Site Water Use}}{\text{ICT Equipment Energy}} \tag{3}$$

The Green Grid believes that, at a minimum, both source- and site-based WUE metrics are beneficial, however these are limited by the lack of calculated and reported annual source energy water usage [9, 12].

2.4 Transparency

A recurrent limitation that researchers and organisations face when developing energy-efficient solutions for data centers is the lack of transparency in data sharing. The Perspective of Hankendi et al. addresses the data gaps that hinder advancements in carbon-aware computing by synthesizing existing literature, proposing a 10-year plan to guide academia, industry, and policymakers toward a more transparent and ethical system [5]. The data gaps comprise unreported operational data (power usage and real-time emissions), embodied emissions (hardware manufacturing and recycling), and external impacts (water usage and community effects).

The social impacts of data centers are particularly difficult to assess, creating a growing need to study the implications of energy-aware strategies on local communities' water and land usage. Data centers are often located in regions chosen for their energy availability and land access, which frequently coincide with areas of higher water stress. In Europe, Spain is one of the leading countries hosting data centers due to its renewable energy availability. The region of Aragon houses several large data centers, one of which was reported to consume up to 500 million liters of drinking water per year [7]. It has been a common practice that DC operators rely on potable water as a direct cooling source [9].

Industry leaders such as Google, Meta, and Amazon have only recently begun releasing high-level sustainability reports, disclosing aggregate Scope 1–3 emissions and efficiency metrics. However, these reports remain insufficient as they lack granularity, for example, omitting pre-inference energy use or not accounting for carbon metrics related to AI hardware. Moreover, reports consolidate multiple factors into single metrics and fail to follow standardised reporting formats, limiting their utility for research. Major reasons leading operators not to disclose hardware configurations or efficiency measures are competitive sensitivities and security concerns [5].

On the topic of water transparency specifically, Digital Realty is one of the few companies publishing a breakdown of water sources [9], and Facebook remains one of the only operators disclosing Water Usage Effectiveness (WUE) figures for its facilities. Despite this, fewer than a third of data center operators track any water metrics at all. These transparency gaps have significant consequences for research quality. Without granular data, researchers are forced to rely on averaged models and estimates that risk being inaccurate [5].

2.5 Energy-aware strategies

Many works have focused on the use of various energy-aware techniques aimed at reducing the power consumption of large-scale CDCs while maintaining performance and Quality of Service (QoS).

Dynamic Voltage and Frequency Scaling. DVFS is a hardware-level technique that reduces the clock speed and voltage of under-utilized CPUs based on workload intensities. This approach is also effective when combined with intelligent prediction algorithms. [1]

Workload Placement Techniques. Virtual Machine (VM) consolidation is another branch of energy-efficient algorithms that determine optimal placements, migrating VMs to fewer physical nodes during low-demand periods, turning off or hibernating underutilized nodes. These techniques aim to achieve a better utilization of the available hardware resources while minimizing energy consumption. Server consolidation entails running VMs on a single physical server.

Job Scheduling Techniques. are algorithms that dynamically allocate tasks to servers in a way that reduces hotspots and evens out the energy demand. Peak avoidance delay non-critical workloads during peak grid carbon intensity. On the other hand, temporal workload shifting, shifts flexible jobs to periods of low carbon intensity. The present thesis will specifically focus on the joint optimization of low carbon and water footprints.

Artificial Intelligence Techniques. AI and Deep learning are being increasingly integrated into energy-aware techniques by learning patterns in workload demands, environmental conditions and user behavior to make proactive decisions. These systems can forecast peak useage times, anticipate cooling requirements and dynamically adjust resource allocations to align with energy-saving goals.

3 Datasets Descriptions

IFCA Cloud. The IFCA Cloud energy consumption metrics dataset was published in May 2025 by Jaime Iglesias Blanco and Alvaro López García [2]. The data was collected from the University of Cantabria, Spain, for a span of four months, from the 14th of December 2024 to the 13th of April 2025. The dataset is composed of two metric files and two auxiliary files. The metric files contain readings of energy consumption of the physical nodes and of the virtual machines running in the IFCA Cloud infrastructure. The energy consumption metrics for the physical nodes were gathered from the Intelligent Platform Management Interface (IPMI), the Running Average Power interface, and from the Scaphandre energy consumption meteorology agent [2]. The data from the virtual machines is collected only using the Scaphandre agent. The two auxiliary files contain the information for both the physical node groups and the virtual machines, including their hardware characteristics. All identifiers had already been anonymized to protect the privacy of users and projects.

ENTSO-E. The European Network of Transmission System Operators for Electricity (ENTSO-E) provides free access to pan-European electricity market and system data through the ENTSO-E Transparency Platform [3]. The platform is operated by ENTSO- E Association on behalf the Transmission System Operators (TSOs) of 36 European countries. The data are composed of Actual Generation per Production Type, and Cross-Border Physical Flows.

The Actual Generation per Production type refers to the net electricity generated by different sources, expressed as average power (MW). The data is provided as an average of instantaneous generation values over, usually, fifteen minutes and is published no later than one hour after the operational period. Cross-Border Physical Flows refers to the flows between bidding zones per market time unit as closely as possible to real time, and no later than one hour after the relevant period. Physical flow is defined as the measured power between neighbouring bidding zones and reflects the economic segmentation of the electricity market. ENTSO-E does not include data from the United Kingdom, but this gap is solved by Wattnet.

Wattnet. Wattnet is an open-source service for tracking the environmental footprint of electricity across Europe. It provides access to real-time, historical, and forecasted indicators on the carbon and water impact of electricity consumption. The tool is specifically designed for researchers, developers, and decision-makers who rely on accurate, transparent, and reproducible environmental metrics. Wattnet will be used in conjunction with the ENTSO-E dataset, through their Python client for the ENTSO-E API.

4 Methodology

The methodology is structured in five parts. The first concerns the construction of the dataset through the integration of three data sources. The remaining four sections each directly address one of the sub-questions, following the order in which they are presented in Section 1.

4.1 Dataset Construction

The primary dataset used in this thesis is the IFCA Cloud energy consumption dataset, which contains node- and VM-level power readings collected between December 2024 and April 2025. To compute carbon and water footprint signals, the IFCA dataset will be augmented using two sources: the ENTSO-E Transparency Platform and the Wattnet API. ENTSO-E provides hourly electricity generation mix and cross-border flow data for Spain, while Wattnet will account for electricity imports and exports across neighboring grids through their flow-tracing methodology. The combination of these sources aims to create a unified time-series dataset at node and VM granularity, with associated CF and WF signals aligned by timestamp. Explanatory Data Analysis (EDA) will be applied to all sources beforehand, while the resulting dataset will be used to answer the research questions presented in this Thesis.

4.2 Energy Consumption Forecasting

To answer RQ1, different classes of interpretable machine learning models will be trained to forecast node- and VM-level energy consumption from operational features. In terms of baseline interpretable models, a Ridge regression model will provide a linear lower bound, enabling direct coefficient-based feature attribution.

A Random Forest and XGBoost model will then be evaluated as non-linear but still interpretable alternatives, with feature importance scores computed via SHAP values to ensure explainability. Finally, an LSTM and GRU model will be trained to capture temporal dependencies in the energy signal. For each model a train/validation/test split will be used to prevent data leakage. Models will be evaluated using RMSE and MAE on the final month of data held out as the test set. The forecasting outputs of the best-performing interpretable model are used as input to the strategy simulation in Section 4.3.

4.3 Strategy Simulation and Sustainable Impact

To answer RQ2, energy-aware operational strategies (DVFS, workload placement and job scheduling techniques) will be simulated retrospectively on the IFCA Cloud dataset. Since live experimentation on the IFCA infrastructure is not feasible, each strategy will be systematically simulated applying a set of rules, that mimic the type of energy-aware strategy, on the historical workload trace from the original dataset. For each simulated strategy, total energy consumption, CF reduction, and WF reduction are computed relative to the unmodified baseline trace retrieved from the reported results in the original IFCA Cloud Dataset. CF and WF are calculated by multiplying the per-timestep energy consumption with the corresponding Wattnet intensity factors. Combinations of strategies are also evaluated to identify whether they are complementary or conflicting in their sustainability outcomes.

4.4 User Impact and Fairness Analysis

To answer RQ3, the human impact of each simulated strategy is assessed in terms of workload delays and distributional fairness across user groups. Workload delay is estimated as the difference in job completion time between the simulated and baseline traces, aggregated per job and per user group. Resource availability is assessed through VM startup wait times and rejection rates under each strategy. User groups will be identified through further exploratory data analysis of the anonymized project and user identifiers in the IFCA Cloud dataset. For each strategy, the distribution of delays across groups is analyzed using the Gini coefficient as a measure of inequality, supplemented by group-level comparisons of mean and variance in delay. This will enhance the explainability of the models and results while also allowing for the identification of whether any strategy imposes a disproportionate burden on specific user classes.

4.5 Transparency and Decision Support Framework

To answer RQ4, the outputs of Sections 4.3 and 4.4 are synthesized into a structured trade-off analysis that quantifies, for each strategy, the relationship between sustainability gains and user impact costs. SHAP-based explanations from the forecasting model are incorporated to provide feature-level transparency on which operational conditions drive energy consumption. The result will serve as a decision support output intended for cloud operators.

5 Risk Assessment

The identified risks of the proposed project can be divided into data availability and methodology.

The first risk regards the limited temporal scope of the IFCA Cloud dataset, which covers only four months (December 2025 to April 2025). This time frame can pose a constraint to the generalization of the forecasting models. As this risk cannot be avoided, it will be explicitly acknowledged as a boundary in the final submission.

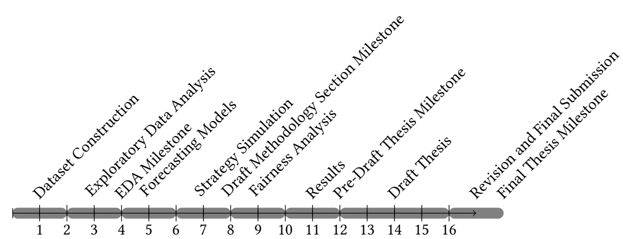
Another risk comes from the anonymization of user groups used by the IFCA Cloud dataset. If the granularity of these identifiers is insufficient to form meaningful user groups, the fairness analysis in RQ3 may yield inconclusive results. The EDA milestone will also serve to identify whether this is an issue and, if necessary, RQ3 will be reframed.

The last identified risk posed by the IFCA Cloud dataset is the absence of ground truth for the energy-aware strategy outcomes. We don't have access to actual CF and WF reductions achieved after applying any strategy. Thus, validation will have to rely on comparison with projection ranges from the published literature discussed in Section 2.

On the topic of energy-aware strategies, we arrive at the risks of the proposed methodology in Section 4.3. Since live intervention on the IFCA infrastructure is not feasible, all energy-aware strategies are simulated. This means that simulated outcomes depend on the realism of the rule-based approximations. To address this, assumptions will be documented transparently, and results will be framed as indicative bounds rather than precise operational predictions.

6 Project Plan

This thesis plan begins from the week of the deadline of the Thesis Design, the 23rd of February 2026, and continues until the final thesis submission deadline on the 26 of June 2026. Each subsection of the methodology described in Section 4 is grouped with the set milestones expected from the DS Thesis plan. Thereby, the dataset aggregation and exploratory data analysis will be completed before the EDA milestone on the 23rd of March. The forecasting and energy-aware strategy simulations from the methodology will have to be concluded before the draft methodology milestone on the 20th of April, while the fairness analysis on the results will be conducted prior the pre-draft Thesis milestone of the 25th of May. Revisions, discussions, limitations, future work and conclusions will mostly be brought out before the draft and final Thesis deadline of the 15 and 26 of June, respectively.



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